

24 jan 2024

① 3 REGEX that match any line that ~~and~~ contains at least 2 vowels, but no digits

1) $^{\wedge}[\wedge 0-9]^*[aeiouAEIOU][\wedge 0-9]^*[aeiouAEIOU][\wedge 0-9]^* \$$
 ↓ non-digits ↓ one vowel ↓ non-digit ↓ one vowel ↓ non-digit ↓ 0 or more times
 0 or more times 0 or more times

2) $^{\wedge}[\wedge [:digit:]]^*[aeiouAEIOU][\wedge [:digit:]]^*[aeiouAEIOU][\wedge [:digit:]]^* \$$
 ↓ non-digit (POSIX class) ↓ non-digit ↓ non-digit

3) `grep -E '^[\w 0-9]^*[aeiouAEIOU][\w 0-9]^*[aeiouAEIOU][\w 0-9]^* $' file`

② GREP cmd displaying all lines in a file that contain an even nr. of vowels.

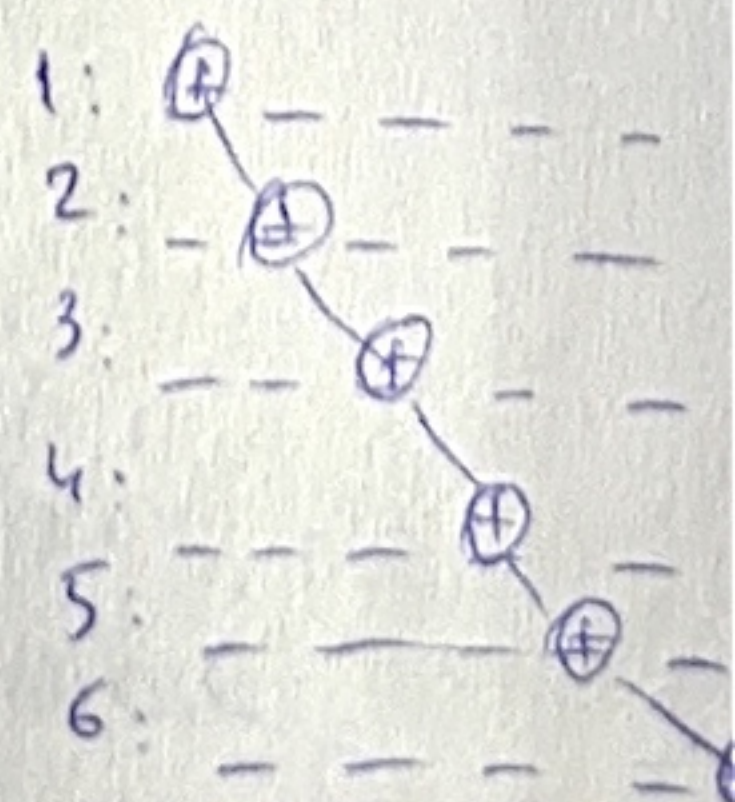
`grep -Ei '^( [\w^aeiou]^* [aeiou] [\w^aeiou]^* [aeiou] )^* [\w^aeiou]^* $' filename.`
 ↓ non-vowel ↓ vowel ↓ vowel

③ SED cmd displaying from a file only ^{the} lines containing exclusively a valid arithmetical expression of integer additions and subtractions.

beginning
`sed -n -E '/^[\w+-]?[0-9]+([\w+-][0-9]+)*$/p' filename`
 ↓ optional sign (+ or -) ↓ integer ↓ repeated 0 or more times ↓ print end
don't print lines automatically

④ AWK → sum of ~~current~~ fields of ~~the line~~ equal to the line index.

`awk '{if (NR <= NF) S+=$NR} END {print S}' filename`



⑤ 2 solutions for hiding a command's stdout and stderr by redirecting them to /dev/null.

1) `command > /dev/null 2>&1`
stderr ← file descriptor → same place as stdout
redirect

2) `command &> /dev/null`
bash shortcut

⑥ UNIX shell script that asks for directory name and insists until it gets a directory that does not exist already.

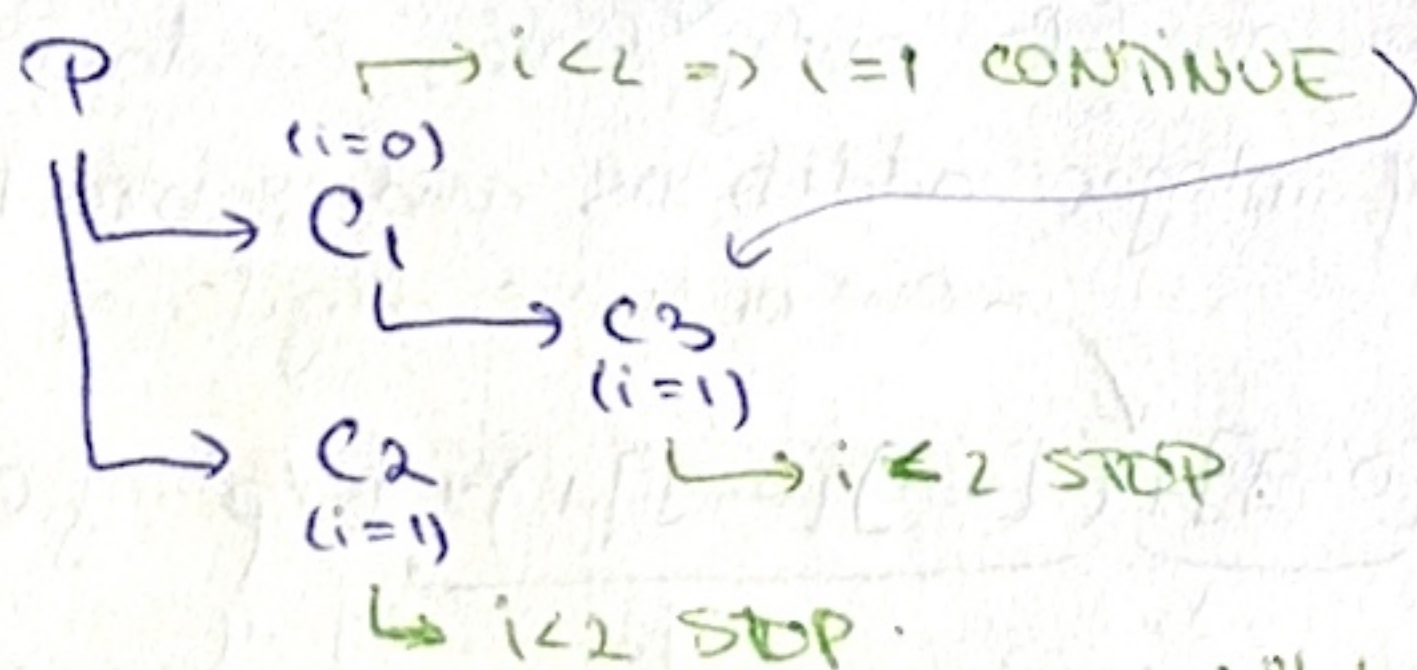
```
#!/bin/bash
while true; do
    read -p "Directory: " d
    if [ ! -d "$d" ]; then
        break
    fi
done
```

check if the path stored in the variable `d` exists and is a directory.

⑦ Process hierarchy:

```
for (i=0; i<2; i++) {
    fork();
    exec("/etc", "/etc", NULL);
}
```

directory => not executable => fails. => loop continues like there is no exec =>



⑧ overwrite content of file b.txt

```
int fd = open("b.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
dup2(fd, STDOUT_FILENO);
close(fd);
execp("sort", "sort", "a.txt", NULL)
```

if "b.txt" does not exist, create it.

overwrite current content

OWNER: R, W
GROUP: R
OTHERS: R

R+W

executable program

first argument passed to the sort cmd

Not advisable to communicate bidirectionally via pipe.

We need to create two pipes for bidirectional comm. because PIPE IS A ONE BYTE STREAM, therefore $A \rightarrow B$ one direction, $B \rightarrow A$ another direction. Without two pipes, deadlocks and race conditions might appear.

(see 19 June 2023 for FIFOs)

⑩ What ^{will} display? Nothing, because no process has opened fifo for reading, so cat f is never reached.

```
mkfifo f
echo asdf > f
cat f
```

⑪ Process over thread? (see 19 June 2023)

when we need: - separate memory (reduce shared memory corruption)

- security!

- distribution to multiple machines

- to minimise crashes (a crash in one process does not affect the outcome of the other)

⑫ Critical resource (see 19 June '23)

= shared resource that can be accessed / modified by multiple threads / processes and must be protected via mutual exclusion (shared var, file, pipe, buffer)

⑬ Why does the pthread_cond_wait call get a mutex as an arg.?

! When pthread_cond_wait is called, the process/thread ⁽¹⁾ releases (unlocks) the associated mutex and blocks itself (sleeps) ⁽²⁾ until another process/thread signals the condition with pthread_cond_signal (wake) ⁽³⁾, after which the process/thread reacquires the mutex ⁽⁴⁾ and gets back to the running state.

• it receives the mutex as an argument because it must atomically:

unlock mutex ⁽¹⁾ → sleep ⁽²⁾ → wake ⁽³⁾ → lock mutex ⁽⁴⁾

⇒ this lets other threads change the condition while the thread sleeps.

⑭ pthread_mutex_lock ↔ pthread_rwlock_wrlock

only one thread allowed to enter

only one writer allowed to enter.

⇒ they are both exclusive ⇒ should behave the same, but rwlocks do more work.

⑮ sem_wait on semaphore initialised with 0?

! $\Rightarrow$ the calling thread blocks until another process thread calls: sem_post(&sem)

⑯ How to decrement the value of a POSIX semaphore? (see 13, 12 jun 2026) ^{ex. (9/7)}

sem_wait(&sem)

$\rightarrow$ counter decreases when a thread enters. $\Rightarrow -1 \Rightarrow$ blocks

$\rightarrow$ counter increases when a thread leaves

⑰ How to prevent deadlocks?

- no-presumption
- consistent locking order
 - avoid nested locks
 - keep critical sections short
 - trylocks / timeouts if needed

⑱ State transition of a process when writing to a file.

- !
- If write blocks $\Rightarrow$ running state $\Rightarrow$ blocked state (waiting) $\Rightarrow$ ready state $\Rightarrow$ running
 - If regular cached files $\Rightarrow$ may return quickly

?
same as
pthread_cond_wait

⑲ Superblock on a Linux disk.

SUPERBLOCK = stores filesystem metadata

$\rightarrow$ nr. of inodes

$\rightarrow$ nr. of zones

$\rightarrow$ inode allocation bitmap

$\rightarrow$ free-space bitmap

$\rightarrow$ zone size

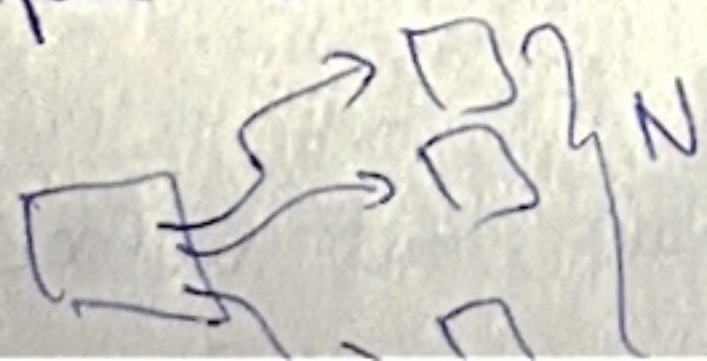
$\rightarrow$ filesystem state

inode = !

⑳ Considering that a block can contain N addresses towards other blocks, how many data blocks are addressed by an i-node's double and triple indirections together?

- double indirection addresses: $N * N = N^2$ data blocks.

- triple indirection addresses: $N * N * N = N^3$ data blocks



$N^2 + N^3$ data blocks together (+)